Predicting game completion time

Can metadata help players choose a game that fits their time budget?

Target: IGDB main-story hours (regression)

Syrym Shadiyarbek and Yerdaulet Akhanov

Machine Learning Algorithms Endterm, SE2424

The same games now have richer metadata

5,229 games

39 predictor fields

New information	Observed games
IGDB themes	4,749 / 5,229
IGDB game modes	5,092 / 5,229
Steam user tags	3,333 / 5,229
Current US price	2,861 / 5,229

Original year and genre features remain. New descriptors add gameplay detail.

Missing observations stay missing. Prices describe current listings.

Training folds control preprocessing and selection

4,183 training games

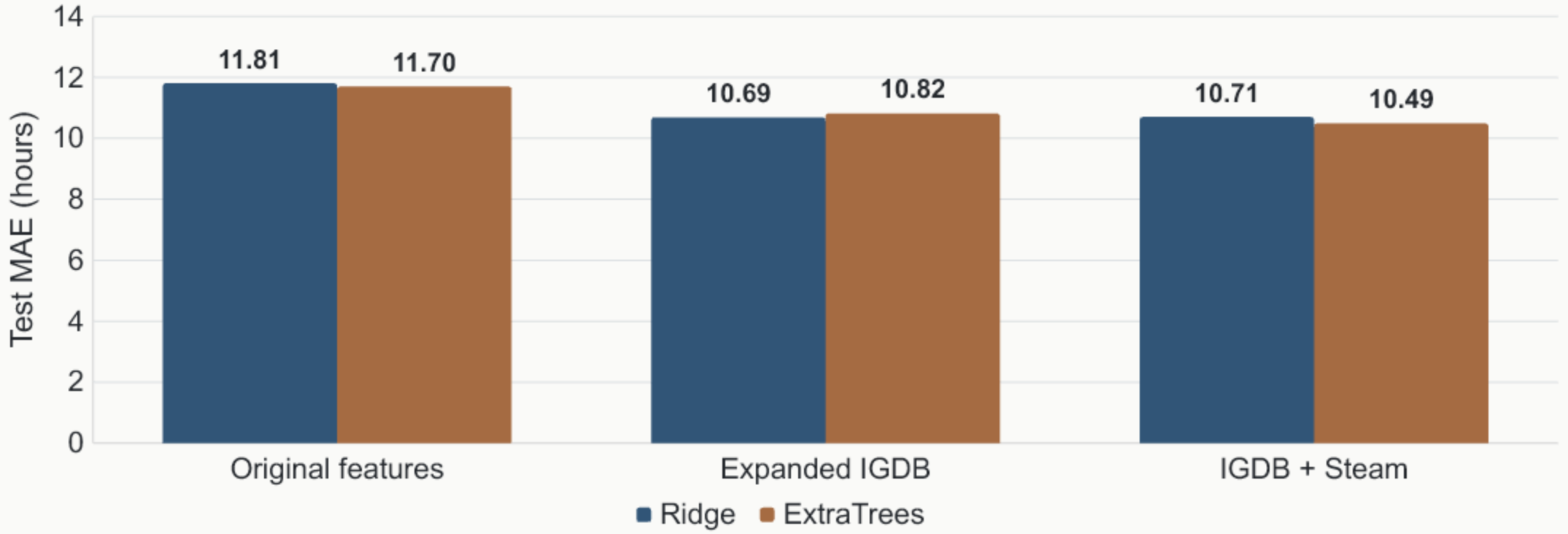
1,046 test games

Stage	What happens
Three training folds	Fit encoders, imputation, scaling and models
Validation folds	Compare settings with MAE in hours
Select and refit	Choose lowest mean CV MAE, then refit training data
Saved test set	Evaluate all models on the same held-out games

IDs, descriptions, alternate completion times and duration tags are excluded.

Fixed seed 42. The previously seen test supports a stage comparison.

Additional metadata reduces test error



Estimator settings stay fixed within each series. Lower MAE is better.

Steam helps ExtraTrees here. Its additional gain for Ridge is unclear.

Random forest has the lowest training-CV MAE

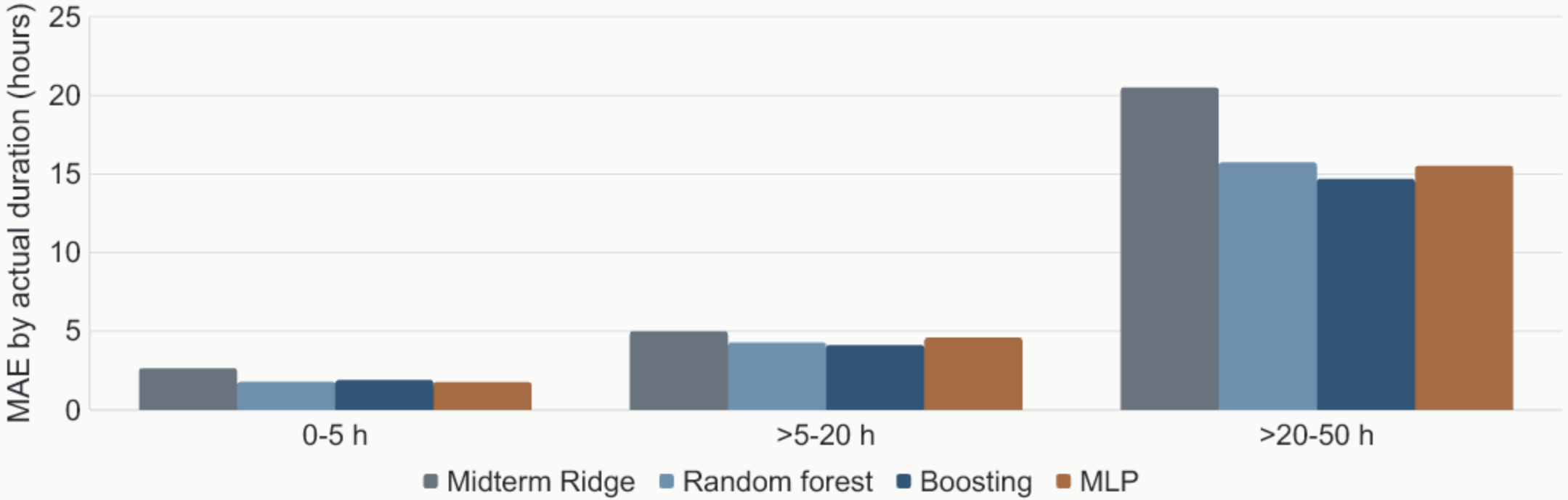
Model	CV MAE ± SD (h)	Test MAE (h)	Test R^2
Median baseline	11.075 ± 0.716	13.01	-0.041
Ridge original (refit)	10.074 ± 0.680	11.81	-0.012
Ridge expanded	9.157 ± 0.680	10.71	0.024
ExtraTrees expanded	8.772 ± 0.743	10.49	0.022
Tuned random forest	8.636 ± 0.642	10.37	0.022
Tuned boosting	8.639 ± 0.602	10.26	0.023
Tuned MLP	9.016 ± 0.637	10.41	0.036

RF: 8 settings. Boosting: 8 settings. MLP: 4 settings.

Selected by training CV: Random forest (near tie)

Complete 13-model comparison: report and notebook.

Errors differ across game duration ranges

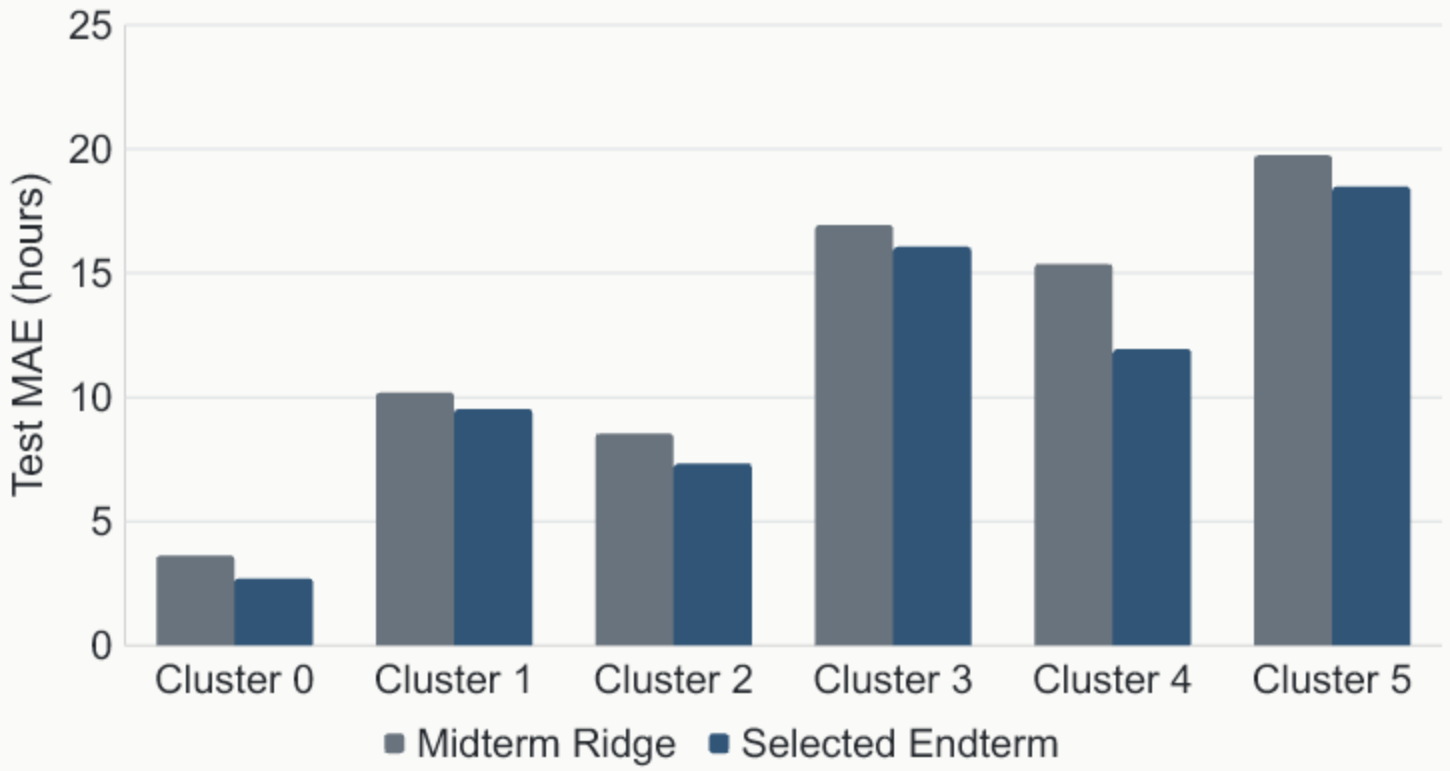


RF vs boosting: RF leads at 0-5 h. Boosting leads at >5-20 h.

13 games above 100 h account for 44.8% of selected-model absolute error.

Chart: 1,004 test games up to 50 h. Overall results retain all 1,046 test games.

Clusters reveal where Midterm Ridge fails



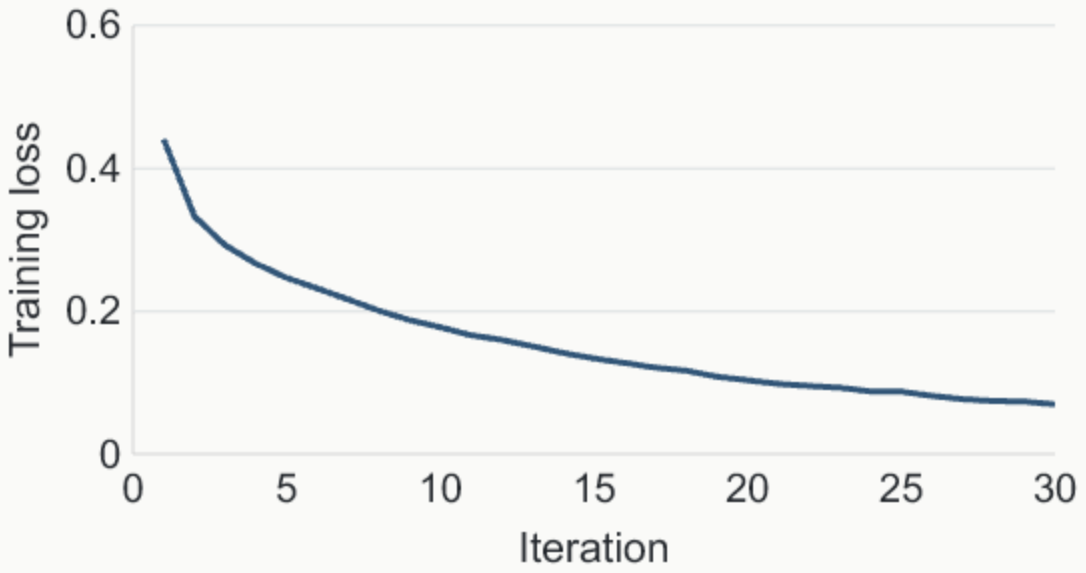
65 PCA components
85.1% variance retained

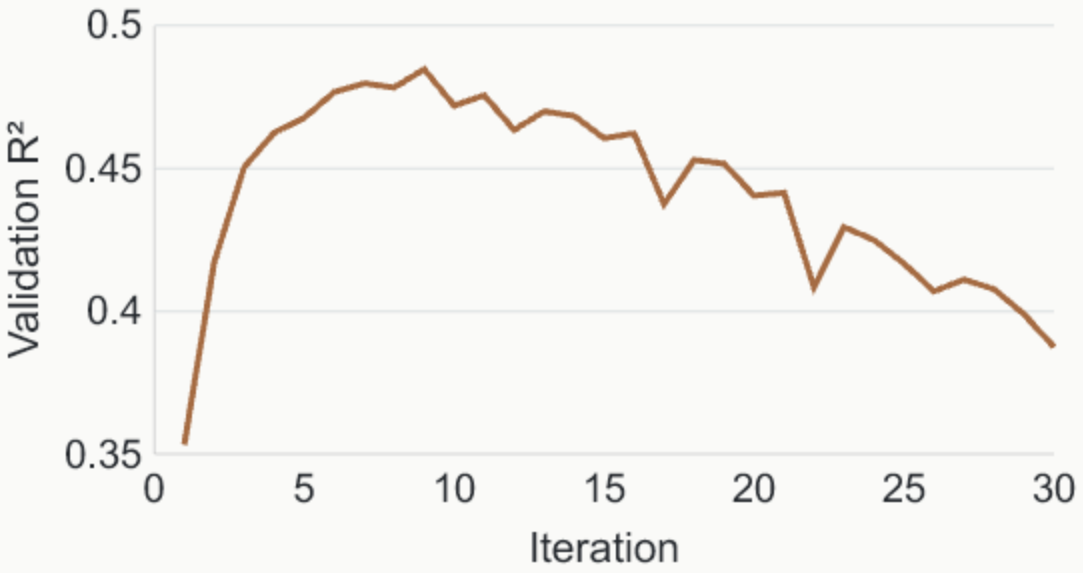
k = 6
Silhouette 0.078
Seed ARI 0.970

The Novelist (cluster 0): Drama, Stealth, Single player.

Actual 2.05 h. Midterm 4.95 h. Endterm 2.87 h. Groups overlap (silhouette 0.078).

MLP validation peaks before training stops





Hidden layers 64 / 32. L2 0.1. Best validation at iteration 9.

Training stops at 30 iterations and restores the best validation weights.

MLP test MAE 10.41 h. Selected model 10.37 h.

Training loss uses standardized log targets. Validation R² uses the same transformed target.

Conclusions and the next experiment

What improved

Test MAE is 10.37 h, improving on Midterm Ridge by 1.45 h.

Median absolute error is 2.26 h.
The model supports rough time-budget estimates.

Richer IGDB features help both controlled comparisons. Steam adds value for ExtraTrees.

Limits and Final test

Long and open-ended games remain difficult. Current prices, selective Steam coverage and a reused test limit generalization.

Use IGDB descriptions linked by game ID. Compare structured, TF-IDF text and combined inputs with fold-fitted features and the same CV MAE.

Select settings by training CV, then evaluate the locked comparison.

Official IGDB API, with a terms check before refresh.